BIOCHEMISTRY CUMULATIVE EXAM

April 8, 1995

Redox potentials are commonly defined according to the Nernst Equation with the midpoint potential, Em, being that potential at which oxidized and reduced species are present in equal amounts.

$$E = E^0 + 0.06 \log([ox]/[red]) \quad \text{for 1 electron transfers}$$

In many biological systems, a change in oxidation state is sometimes accompanied by gain or loss of a proton, dependent of course on the pH of the system. In other words there is a coupling between redox and protonation states. The group gaining or losing the proton need not be the redox center but can be another ionizable group on the protein.

Consider the following system:

$$A_{ox} + e^- \xleftrightarrow{E^0} \begin{array}{c} A_{red} \\ \rightleftarrows H^+ \\ HA_{red} \end{array}$$

where reduction of A can result in either A_{red} or HA_{red}, depending on the pH of the medium and the pKa of the group being protonated. The pKa is defined in terms of the reaction

$$HA_{red} \xleftrightarrow{} H^+ + A_{red} \qquad Ka = [H^+][A_{red}]/[HA_{red}]$$

a. (25 points) Beginning with the Nernst Equation, show that the dependence of midpoint potential (Em) on $[H^+]$ and Ka for this system will be described by the following equation.

$$Em = E^0 + 0.06 \log(1 + [H^+]/Ka)$$

(Hint: You may assume that in the Nernst Equation "ox" and "red" refer to the sum of all species in the indicated redox state.)

b. (25 points) Beginning with the equation you have derived in a., what would you expect the relationship between Em and pH to be under conditions where $[H^+] < Ka$? No credit given unless you explain your answer.

c. (25 points) Beginning with the equation you have derived in a., show that the relationship between Em and pH under conditions where $[H^+] > Ka$ can be approximated by

$$Em = E^0 + 0.06\,[pKa - pH]$$

d. (25 points) Assuming that we are dealing with a protein which serves as a one electron carrier and has a redox-linked ionizable group having a pKa of 6.0, sketch a plot of how you would expect the Em to vary as a function of pH over the range 2-10. Be sure to include in your sketch the expected slopes of any expected linear segments of your plot.